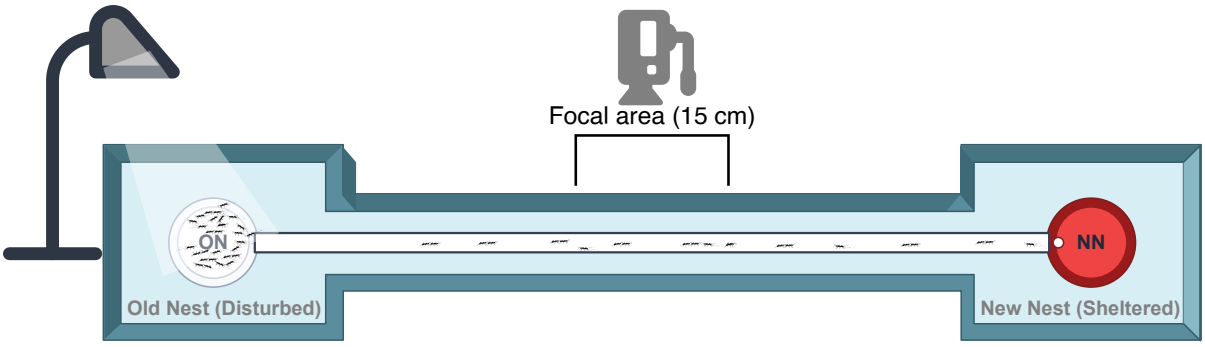
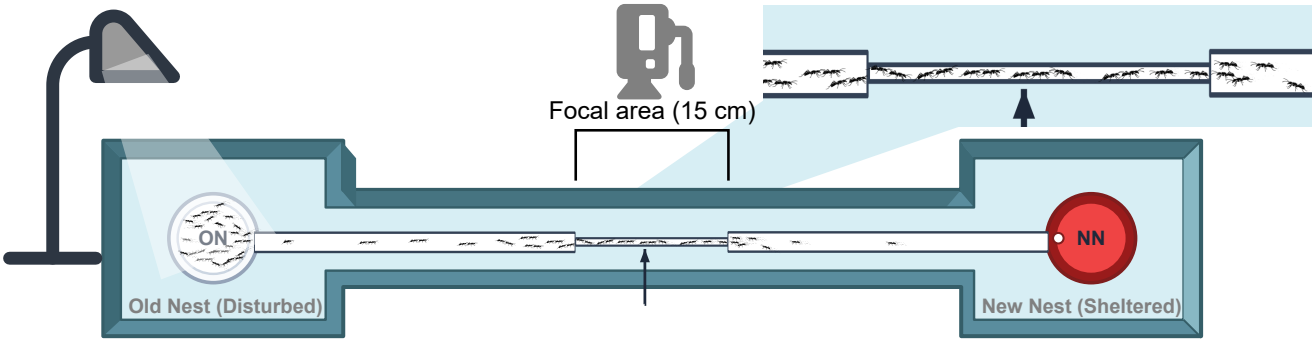
Experimental setup



a. Control: Multi-lane path (75 cm × 2.5 cm wide, 5 lanes; unconstrained bidirectional counterflow)

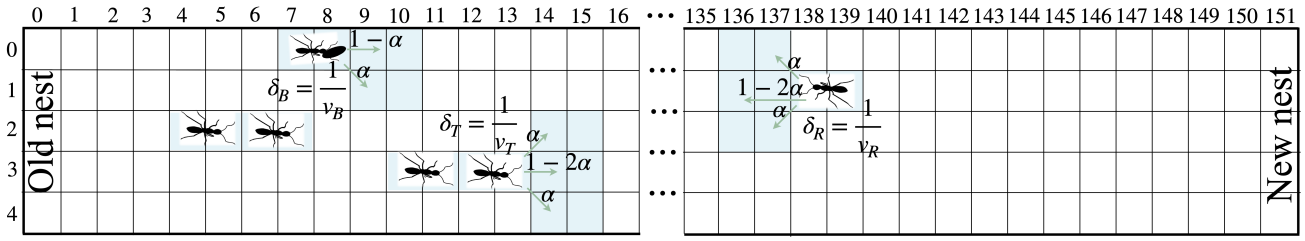


b. Manipulation: One-lane bottleneck path (75 cm total; 15 cm × 0.5 cm constriction flanked by 2.5 cm approaches)

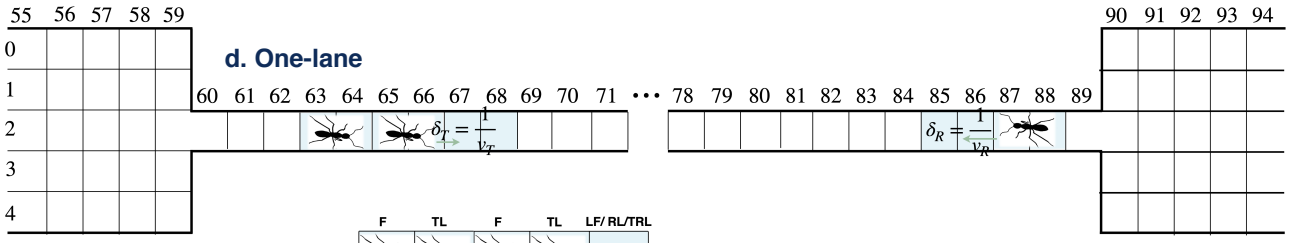
Nest relocation direction

Agent Based Model

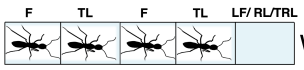
c. Multi-lane



d. One-lane

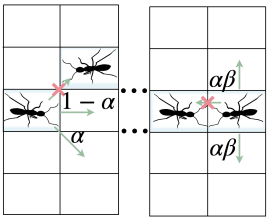


e. Jam



Waits more than 4s

f. Restriction



g. Interactions

